



Tracing a TCP/IP Communication: Opening a Secure Website (HTTPS)

A methodical layer-by-layer architectural analysis: *"How does a single click become a secure web page?"*

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The TCP/IP Model: Four Layers, One Job Each

Application Layer

PDU: Data**"Prepares the message"**

Constructs the HTTP GET request, initiates TLS 1.3 session encryption, and runs DNS name lookups.

Transport Layer

PDU: Segment**"Gets it there reliably (or fast)"**

Manages process ports (Source: 51342, Dest: 443), connection state, sequencing, and retransmission.

Internet Layer

PDU: Packet**"Finds the destination network"**

Attaches logical IP addresses (Src: 192.168.1.10, Dst: 93.184.216.34) enabling cross-network internet routing.

Network Access Layer

PDU: Frame**"Moves the bits physically"**

Encapsulates packets into Ethernet frames with physical MAC addresses across the local cable/Wi-Fi link.

ARCHITECTURAL MAPPING

Alignment with OSI 7-Layer Model

Layers 7, 6, 5: Application, Pres., Session**Application****Layer 4:** Transport**Transport****Layer 3:** Network**Internet****Layers 2 & 1:** Data Link & Physical**Network Access**

Key Distinction: The TCP/IP model integrates session handling and presentation syntax directly into the Application layer, creating an efficient, implementation-driven networking architecture.



1. DNS Resolution

A DNS query is transmitted using **UDP Port 53** to resolve human-readable domains (e.g., portal.company.com) into numerical IP addresses before any connection begins.

PDU: DNS Query Message
Transport: UDP / 53



2. HTTP GET Construction

The browser formats an HTTP/1.1 or HTTP/2 GET request string, assembling request headers, Host identifier, user-agent parameters, and expected MIME types.

PDU: Application Data
Content: Plaintext HTTP GET



3. TLS 1.3 Encryption

Before leaving the client, the application data is handed over to **TLS 1.3** for symmetric cryptographic encryption, transforming plaintext HTTP into secure HTTPS.

Destination Port: 443
Payload: Encrypted TLS Record

Transport & Internet Layers: Session & Routing

CONNECTION SETUP

TCP 3-Way Handshake (Port 443)

1. SYN (Synchronize)

Client → Server

Client selects an Initial Sequence Number (ISN) and sends SYN flag set.

2. SYN-ACK

Server → Client

Server acknowledges client ISN (ACK=ISN+1) and sends its own sequence number.

3. ACK (Acknowledge)

Client → Server

Client acknowledges server sequence; reliable socket is ready for TLS exchange.

HEADER ENCAPSULATION

Headers Added During Packaging

TCP Segment (Layer 4)

Transport

Attaches Source Port (ephemeral **51342**) & Destination Port (**443**).

IP Packet (Layer 3)

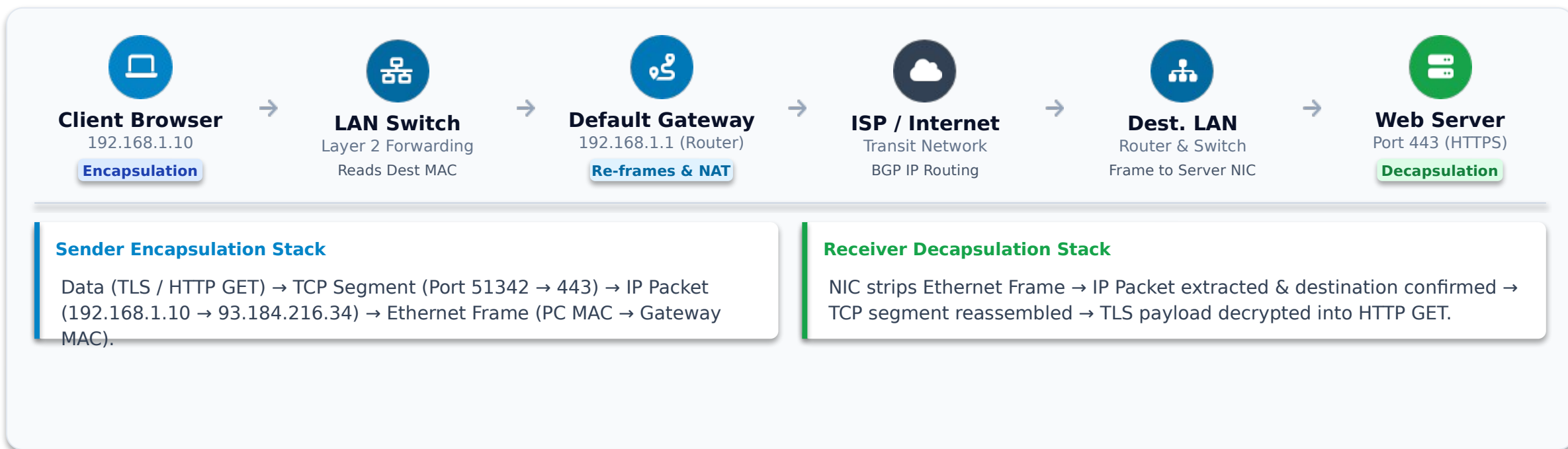
Internet

Attaches Source IP (**192.168.1.10**) & Destination IP (**93.184.216.34**).

Cross-Network Routing: Routers across the internet examine only the Destination IP to forward the packet. The Transport port numbers and TLS payload remain untouched end-to-end.



Original Packet Journey Diagram — Opening a Secure Website



Return Path: The web server's TLS/HTTP response is encapsulated identically in reverse, routed across internet routers, and decapsulated by the client's browser.

LAYER 2 FRAMING

Ethernet / Wi-Fi Framing

The Network Access layer wraps the IP packet into an Ethernet frame containing local physical MAC addresses.

Hop-by-Hop Replacement:

MAC addresses are stripped and rebuilt at every router interface along the path while IP addresses remain constant.

Frame Check Sequence (FCS):

Cyclic Redundancy Check (CRC) validates that physical signals were not corrupted during physical transmission.

NEXT-HOP MAPPING

Address Resolution Protocol (ARP)

The client PC knows its Gateway IP (192.168.1.1) but requires its physical MAC address to transmit the frame over the wire.

1. ARP Broadcast Request:

"Who has 192.168.1.1? Tell 192.168.1.10" broadcast to all hosts on the LAN (FF:FF:FF:FF:FF:FF).

2. Unicast ARP Reply:

Router responds with its MAC; client caches mapping in its local ARP table for immediate frame dispatch.

Protocol Analysis Table: Opening a Secure Website

Layer	Protocol	PDU	Addressing Info	Purpose
Application	DNS	Data (message)	Domain name → IP address	Resolves website domain name to IP before anything else occurs.
Application	HTTP over TLS (HTTPS)	Data (message)	None (relies on ports below)	Requests the web page and carries the encrypted response back from server.
Transport	TCP	Segment	Src: 51342 / Dst: 443	Reliable, ordered connection; opens session via 3-way handshake.
Internet	IP	Packet	Source IP / Destination IP	Logical addressing & routing — moves packets across internetworks.
Network Access	Ethernet / Wi-Fi	Frame	Source MAC / Destination MAC	Physical delivery across one local link; re-applied at every hop.

**LAYER 3 •**

LOGICAL

IP Address (IPv4)

Routable Globally: Logical, software-assigned identifier that denotes exactly where a device sits on the global internetwork.

Static vs. Dynamic: Manually assigned by a network engineer or dynamically leased on boot via DHCP protocol.

Hierarchical Structure: Split into Network portion and Host portion (e.g. 192.168.1.10 /24).

LAYER 2 •

PHYSICAL

MAC Address (Ethernet)

Flat & Non-Routable: Physical hardware address burned directly into the Network Interface Card (NIC); meaningful only on the local segment.

48-Bit Hex Format: 12 hexadecimal characters separated by colons (e.g. 00:1A:2B:3C:4D:5E).

OUI & Device Identifier: First 24 bits designate the vendor (OUI); remaining 24 bits designate the unique hardware interface.



Port 53

DNS Protocol

Resolves domain names to IP addresses.
Operates primarily over UDP (or TCP for large zone transfers and query responses).

Port 443

HTTPS (HTTP over TLS)

Encrypted, authenticated web traffic delivered over TCP, authenticated via X.509 server TLS certificates.

Port 51342

Client Ephemeral Port

Dynamic client-side port chosen from the private range (49152–65535) to identify the specific browser process socket.

Core Rule: Ports identify the specific *application or service socket* on a host — not the device itself. Finding the target computer is the IP address's job; delivering data to the exact browser tab or server daemon is the port number's job.



Transport Layer: TCP vs UDP Mechanics

TCP Mechanics

Reliable

Layer 4 Scope: Process-to-process communication managed via standardized port numbers.

Connection-Oriented: Opens session via three-way handshake (SYN → SYN-ACK → ACK) before sending data.

Guaranteed Delivery: Segmentation and reassembly with sequence numbers ensure correct ordering.

Error Retransmission: Automatically resends dropped packets upon timeout or missing acknowledgment.

Flow Control: Sliding window mechanisms prevent sender from overwhelming the receiver's buffer.

UDP Mechanics

Connectionless

Instant Transmission: Dispatches datagrams immediately with zero handshake setup or session state.

Best-Effort Delivery: No acknowledgments; relies entirely on the application layer to handle loss.

No Ordering Guarantee: Datagrams may arrive out-of-order or duplicate without protocol detection.

Zero Retransmission: Dropped datagrams are permanently ignored rather than queued.

Minimal Overhead: 8-byte header provides maximum throughput and minimal processing delay.

TCP vs UDP: Comprehensive Comparison Table

Feature	TCP	UDP
Connection	Connection-oriented — 3-way handshake (SYN, SYN-ACK, ACK) before data flows	Connectionless — no handshake, no session; sends immediately
Reliability	Reliable — lost segments are detected and retransmitted	Unreliable — no retransmission; the application must handle loss if it matters
Ordering	Guaranteed — sequence numbers reassemble data in the correct order	Not guaranteed — datagrams may arrive out of order
Flow control	Yes — sliding window prevents overwhelming the receiver	None
Speed / overhead	Slower, higher overhead (headers + handshake + acknowledgments)	Faster, lower overhead (minimal header, no setup)
Data unit (PDU)	Segment	Datagram

Transport Protocol Application Examples

Protocol	Application 1	Application 2	Why this protocol fits
TCP	Web browsing (HTTPS, port 443)	Email (SMTP)	The full page or message must arrive complete and in order — dropped, out-of-order, or scrambled data is completely unacceptable in financial documents or source code.
UDP	Live video/voice calls (e.g. VoIP)	DNS lookups	Speed and freshness matter more than perfect delivery — a late retransmission is worse than a small gap in live audio, and DNS can simply retry if a query packet drops.
TCP Design Priority: Guarantees 100% data integrity and strict packet sequencing over low transmission latency. UDP Design Priority: Guarantees lowest possible latency and minimal processing overhead for real-time streaming.			

**OBSERVED
ERROR****NET::ERR_CERT_DATE_INVALID**

An employee accesses `https://portal.company.com` and the browser blocks entry: *"Your connection is not private"*.

The underlying LAN connection is fully operational — standard web traffic and email continue to function normally.

Diagnostic Goal:

Isolate the exact failing layer methodically using standard networking tools rather than guessing.

1. Physical Link

✓ ip link / ifconfig UP

2. IP Connectivity

✓ ping default gateway OK

3. DNS Resolution

✓ nslookup resolves valid IP

4. Port Reachability

✓ telnet / nc port 443 OPEN

5. TLS Handshake Probe

✗ openssl s_client reports EXPIRED

Remediation:

Re-issue TLS certificate through trusted CA, bind full certificate chain on the web server daemon, reload the service, and verify via OpenSSL.

Authoritative References

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The Transport Layer Security (TLS) Protocol Version 1.3

IETF RFC 793

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